MAST20026 REAL ANALYSIS

Sequences: $\epsilon - M$ Convergence - Solutions

5A

A Note on the Solutions: This solution sheet is intended for you verify correctness of your work during the tutorial. The solutions here do not necessarily meet the standards for exposition expected on assignment submissions.

1. **Proving a Limit: Step -2**

   Each of the sequences below converge. Make a guess to which value of L each sequence converges to.

   (a) $a_n \to 1$

   (b) $b_n \to 0$

   (c) $c_n \to 0$

   (d) $d_n \to 1/3$

   (e) $e_n \to 1$

2. **Proving a Limit: Step -1**

   Sample calculation for (d):

$$\begin{aligned}
\left|\frac{n^2}{3n^2-4} - \frac{1}{3}\right| &< 0.01 \\
\left|n^2 - \frac{3n^2-4}{3}\right| &< 0.01\,(3n^2-4) \\
\frac{4}{3} &< 0.01\,(3n^2-4) \\
\sqrt{\frac{4}{9(0.01)} + \frac{4}{3}} &< n
\end{aligned}$$

3. **Proving a Limit: Step 0**

   Sample calculation for (d):

$$\begin{aligned}
\left|\frac{n^2}{3n^2-4} - \frac{1}{3}\right| &< \epsilon \\
\left|n^2 - \frac{3n^2-4}{3}\right| &< \epsilon(3n^2-4) \\
\frac{4}{3} &< \epsilon(3n^2-4) \\
\sqrt{\frac{4}{9\epsilon} + \frac{4}{3}} &< n
\end{aligned}$$

4. **Proving a Limit: Step 1**

   Let $\epsilon > 0$. Set $M = \left\lceil \sqrt{\frac{4}{9\epsilon} + \frac{4}{3}} \right\rceil$ and let $n > M$. Since $n > M$ we have $n > \sqrt{\frac{4}{9\epsilon} + \frac{4}{3}}$.

   Manipulating, we find:

$$\begin{aligned}
\sqrt{\frac{4}{9\epsilon} + \frac{4}{3}} &< n \\
\frac{4}{3} &< \epsilon(3n^2-4) \\
\left|n^2 - \frac{3n^2-4}{3}\right| &< \epsilon(3n^2-4) \\
\left|\frac{n^2}{3n^2-4} - \frac{1}{3}\right| &< \epsilon
\end{aligned}$$

   Therefore $\left|d_n - \frac{1}{3}\right| < \epsilon$ whenever $n > M$. It follows that $\lim_{n\to\infty} \frac{n^2}{3n^2-4} = \frac{1}{3}$.

5. **Algebra of Limits**

Recall the statement of the Algebra of Limits Theorem

Theorem (Algebra of Limits Theorem)**.** *If $f_n \to \alpha$ and $g_n \to \beta$ are convergent sequences, then*

i. $f_n + g_n \to \alpha + \beta$;

ii. $f_n - g_n \to \alpha - \beta$;

iii. $f_n g_n \to \alpha\beta$; *and*

iv. $f_n/g_n \to \alpha/\beta$ *(provided $\beta \neq 0$ and no term of g_n is 0).*

Using the Algebra of Limits Theorem, determine the limit of each sequence. You may assume that your guesses in question 1 are correct.

(a) Notice

$$\frac{n^2+6}{3n^2-4} = \frac{n^2}{3n^2-4} + \frac{6}{3n^2-4}$$

Let $b_n = \frac{6}{3n^2-4}$ and $c_n = \frac{n^2}{3n^2-4}$. From above we have $b_n \to 0$ and $c_n \to \frac{1}{3}$. Therefore $c_n + b_n \to \frac{1}{3}$. And so

$$\lim_{n\to\infty} \frac{n^2+6}{3n^2-4} = \frac{1}{3}$$

(b) Let $e_n = (1, 1, 1, \dots)$. We have $e_n \to 1$ and $h_n = 3e_n$. Therefore $h_n \to 3 \times 1$. That is, $h_n \to 3$.

6. **Comparisons**

Some of the algebra in Question 1 of this tutorial took a lot of work! Looking at our definition of convergence above, there is nothing that says we must choose the least possible M, but in fact we can choose any M that works.

(a) This can be done by completing the square, you will conclude that in order to guarantee

$$\frac{1}{n^2+6n+11} < \epsilon$$

you need

$$n > \sqrt{\frac{1}{\epsilon} - 2} - 3.$$

Unfortunately, the expression under the square root can be negative (namely, if $\epsilon > 1/2$), and the whole right hand side can be negative even if it is defined (if $1/11 < \epsilon < 1/2$). This will make writing the proof painful - you will have to distinguish cases.

(b) Yes. If $x > y > 0$ are real numbers, then $\frac{1}{x} < \frac{1}{y}$. Apply this with $x = n^2 + 6n + 11$ and $y = n^2$.

(c) $M = \left\lceil \frac{1}{\sqrt{\epsilon}} \right\rceil$

(d) Let $\epsilon > 0$. Let $M = \left\lceil \frac{1}{\sqrt{\epsilon}} \right\rceil$ and let $n > M$. Since $n > M$ we have $n > \frac{1}{\sqrt{\epsilon}}$ Manipulating, we find:

$$\frac{1}{\sqrt{\epsilon}} < n$$
$$\frac{1}{n^2} < \epsilon$$

Since $\frac{1}{n^2} < \epsilon$ and $n^2 < n^2 + 6n + 11$ we have

$$\frac{1}{n^2+6n+11} < \epsilon$$

Since $n > 0$ we have $\left|\frac{1}{n^2+6n+11}\right| = \frac{1}{n^2+6n+11}$. Therefore

$$\left|\frac{1}{n^2+6n+11} - 0\right| < \epsilon$$

Therefore $\left|\frac{1}{n^2+6n+11} - 0\right|$ whenever $n > M$. We conclude that $\lim_{n\to\infty} \frac{1}{n^2+6n+11} = 0$.

(e) It is not always true that $n^2 < n^2 - 6n + 4$. Worse, the expression $n^2 - 6n + 4$ is negative for $n < 6$, so no matter what you attempt, some care will be required. One way to proceed would be to notice that for $n \geq 6$, $(n-6)^2 \leq n^2 - 6n + 4$. You can then take

$$M = \left\lceil \frac{1}{\sqrt{\epsilon}} + 6 \right\rceil$$

in the proof, taking care to note the fact that this means $n > M$ implies $n > 6$:

Let $\epsilon > 0$. Let $M = \left\lceil \frac{1}{\sqrt{\epsilon}} + 6 \right\rceil$ and let $n > M$. Since $n > M$ we have $n > \frac{1}{\sqrt{\epsilon}} + 6$, and in particular, $n - 6 > 0$. Thus,

$$\frac{1}{\sqrt{\epsilon}} + 6 < n$$

implies

$$\frac{1}{(n-6)^2} < \epsilon.$$

Since $n > 6$, $(n-6)^2 < n^2 - 6n + 4$ and therefore

$$\frac{1}{n^2 - 6n + 4} < \frac{1}{(n-6)^2} < \epsilon$$

Again since $n > 6$ we have $\left|\frac{1}{n^2-6n+4}\right| = \frac{1}{n^2-6n+4}$, so it follows that

$$\left|\frac{1}{n^2 - 6n + 4} - 0\right| < \epsilon$$

We have shown that $\left|\frac{1}{n^-6n+4} - 0\right|$ whenever $n > M$. Therefore, $\lim_{n\to\infty} \frac{1}{n^2 - 6n + 4} = 0$.